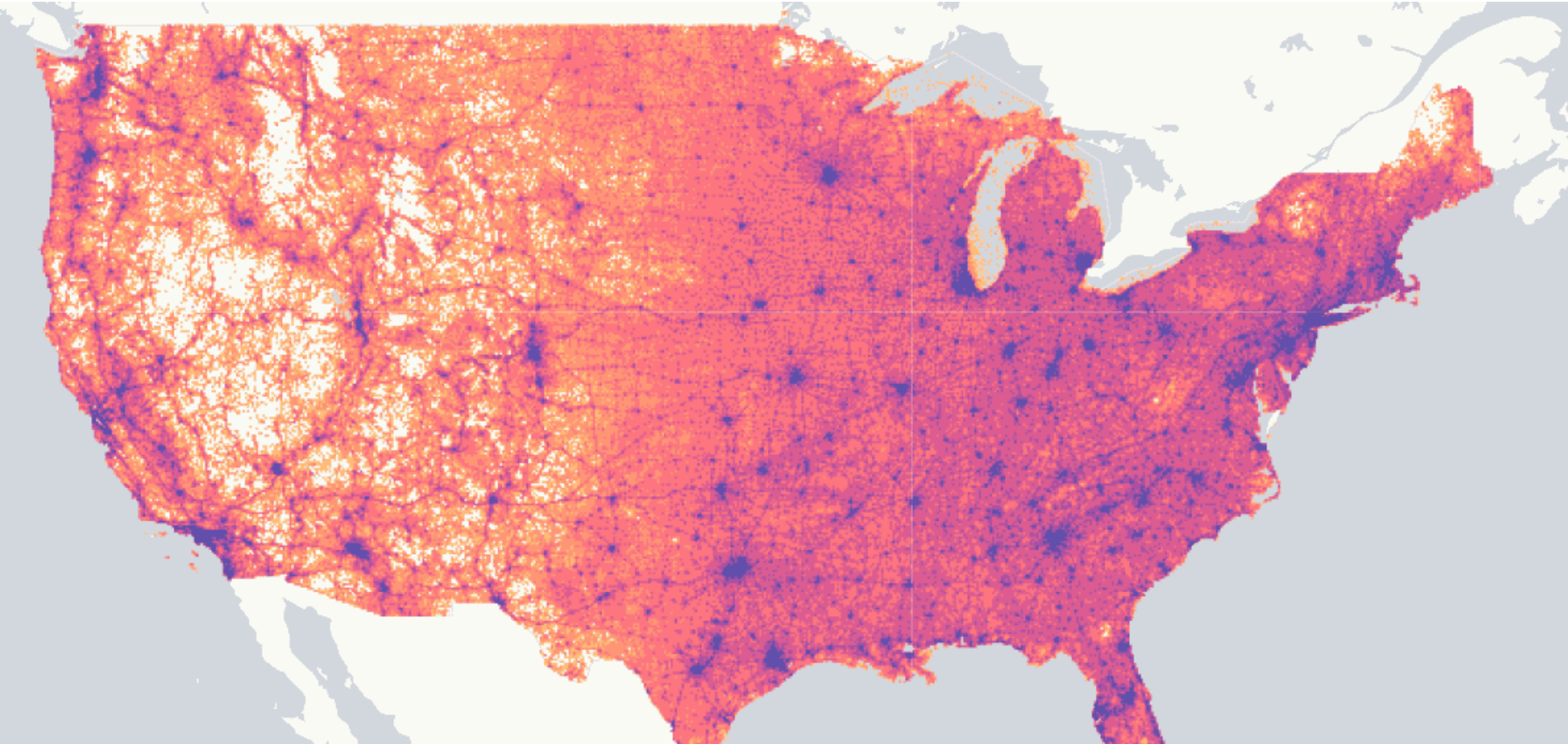
Human flows

Course: Physics of Complex Networks: Structure and Dynamics



Objective: compare data vs human flow models

Gravity Models $\Phi_{ij} = k \frac{N_i^\alpha N_j^\beta}{f(d(i,j))}$

$f(d(i,j)) \propto d^\gamma(i,j)$

$f(d(i,j)) \propto e^{d(i,j)/d_0}$

Radiation Model $\Phi_{ij} = \frac{N_i N_j}{(N_i + S_{ij})(N_i + N_j + S_{ij})}$

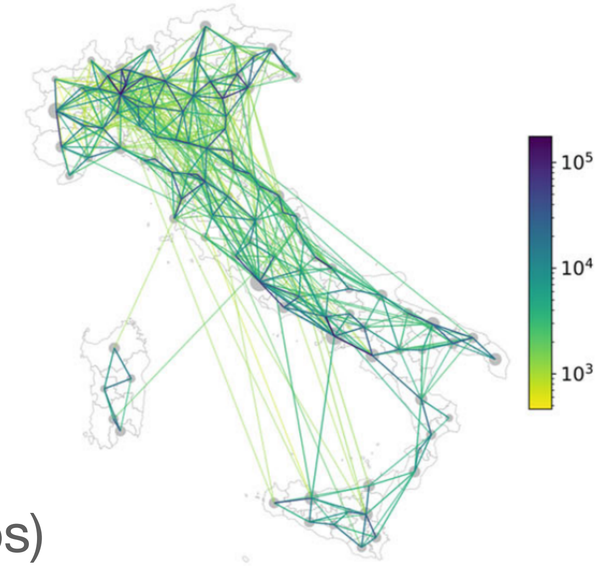
(1) Use population data to build exponential and power-law gravity models. Fit the parameters.

(2) Use population data to build the radiation model. Compare against real data (complete at home if necessary)

Note: in the next session you will use THIS output to run simulations.

(3) Visualize a comparison of data vs gravity vs radiation model (4 panels):

- **Networks**
- **Heatmaps of mobility flows**



Note: The best visualization(s) will be used for the website (and next-year demos)